

Professional Summary

Software Engineer with 4+ years of experience in C++ and Python system-level development. Strong expertise in Windows Internals, multithreading, TCP/UDP networking, and performance optimization. Experienced in building enterprise-grade forensic acquisition systems, memory analysis tools, network monitoring platforms, and high-performance distributed applications.

Core Expertise

- System Programming using Win32 API and NTAPI
- Windows Process, Memory, and Registry Analysis
- Digital Forensics and Incident Response (DFIR)
- Multithreading and Concurrency Architecture
- TCP/IP, UDP, and Real-Time Network Monitoring
- Secure Evidence Acquisition and Integrity Validation
- Performance Optimization and Low-Level Debugging
- Transport Stream Processing (FFMpeg, TSDuck, NDI)

Technical Skills

Languages: C++, Python, C#, JavaScript, SQL

Frameworks/APIs: Win32 API, NTAPI, MFC, Qt5, Django, .NET

Networking: TCP/IP, UDP, Raw Sockets, Packet Monitoring, Network Tracing

Databases: MySQL, SQL Server, SQLite

Tools: Visual Studio, Git, VS Code, SVN, JIRA

Media/Stream Tools: FFMpeg, NDI SDK, TSDuck

Forensic Tools: Volatility3, Memory Artifact Parsing

Professional Experience

Software Engineer — Innefu Labs Pvt. Ltd

Nov 2024 – Present New Delhi, India

- Contributing to the development of RapidFIR (Rapid Digital Forensics & Incident Response) platform.
- Designed and implemented remote endpoint forensic acquisition modules for enterprise environments.
- Developed memory, registry, process, and network artifact extraction components using Windows Internals.
- Engineered structured JSON-based pipelines for real-time artifact parsing and centralized incident analysis.
- Integrated automated triage mechanisms to support rapid threat detection and response workflows.
- Optimized multithreaded data collection modules to minimize system overhead and latency.
- Implemented secure hashing and validation mechanisms to ensure forensic evidence integrity.

Software Engineer — JTEKT Electronics India Pvt. Ltd

May 2024 – Jul 2024 Gurugram, India

- Engineered PLC configuration and visualization software using C++, MFC, and Win32.
- Integrated SQL-driven backend systems to enhance industrial automation reliability.

Software Engineer — Planetcast Media Services Ltd

Aug 2021 – Dec 2023 Noida, India

- Developed high-performance C++ and Python applications for broadcast automation.
- Implemented multithreaded TCP/UDP communication modules for real-time video streaming.
- Integrated FFMpeg, NDI SDK, and TSDuck for transport stream monitoring and processing.
- Improved backend performance by 25% through architectural optimization and concurrency tuning.

Selected Projects

Cipher Sentinel Imager — Python, Qt5

- Developed a comprehensive forensic acquisition solution supporting disk imaging, drive cloning, RAM dumps, and secure wiping.
- Implemented image verification using SHA256 hashing to ensure forensic integrity and admissibility.
- Enabled network-based acquisition from NAS and SAN environments for enterprise-scale investigations.
- Designed optimized I/O pipelines for high-speed acquisition while maintaining data accuracy and reliability.
- Integrated logging, error handling, and audit trail mechanisms for professional forensic workflows.

Network Traffic Tracer with Live Web Dashboard — C++

- Engineered a real-time TCP/UDP network monitoring tool capturing connection metadata, byte transfer metrics, and session duration.
- Designed in-memory structured data storage and JSON-based API endpoints for direct web-based visualization.
- Implemented multithreaded packet handling to ensure minimal packet loss and accurate traffic statistics.
- Built lightweight HTTP interface for live monitoring without intermediate CSV dumping.

Live RAM Extraction Tool — C++

- Designed live memory acquisition utility without generating full system dumps.
- Extracted volatile artifacts including running processes, loaded modules, and active network connections.
- Reduced acquisition time by 60% compared to traditional dump-based approaches.

Windows Prefetch Parser with Web Interface — Python

- Built parser for Windows Prefetch (.pf) files extracting execution metadata and system artifacts.
- Developed web-based dashboard for structured forensic visualization and export functionality.

MultiViewer – Real-Time Broadcast Monitoring System — C++

- Designed multi-source monitoring system handling UDP, RTMP, and transport stream feeds.
- Implemented multithreaded rendering pipeline for simultaneous decoding and display of multiple streams.
- Integrated FFmpeg and TSDuck for transport stream analysis and packet inspection.
- Optimized buffering and synchronization logic to reduce latency and ensure frame stability.
- Built scalable architecture supporting high-throughput broadcast monitoring environments.

Education

Bachelor of Technology (Computer Science & Engineering)

IMS Engineering College, Ghaziabad — 2021

Certifications

Advanced Certificate Programme in Data Science — IIIT Bangalore

Machine Learning with Python — CETPA

MicroPython & MATLAB — NIELIT Calicut